

UVC Camera with DVP

Introduction

This sample code demonstrates how to perform USB camera feature using USB + DVP + OV2640.

Applicable products:

Product series	AT32F435 series
	AT32F437 series

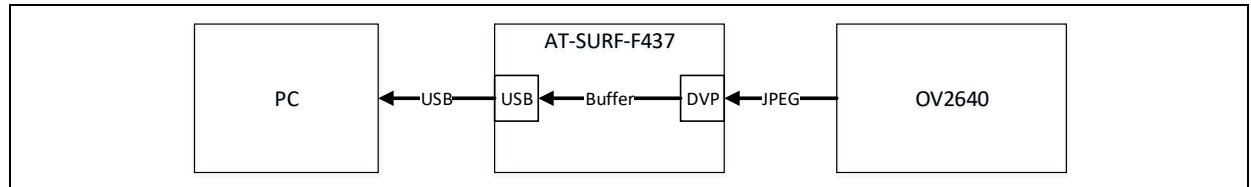
List of major peripherals used:

Peripherals	OTG
	DVP

1 Overview

In the demo, the USB device is enumerated as USB Camera. The DVP+OV2640 is in charge of collecting image data, which are then transferred to the USB host via USB. Currently up to 15 frames can be supported. The OV2640 uses JPEG format, and the resolution is 320x240.

Flow chart below:

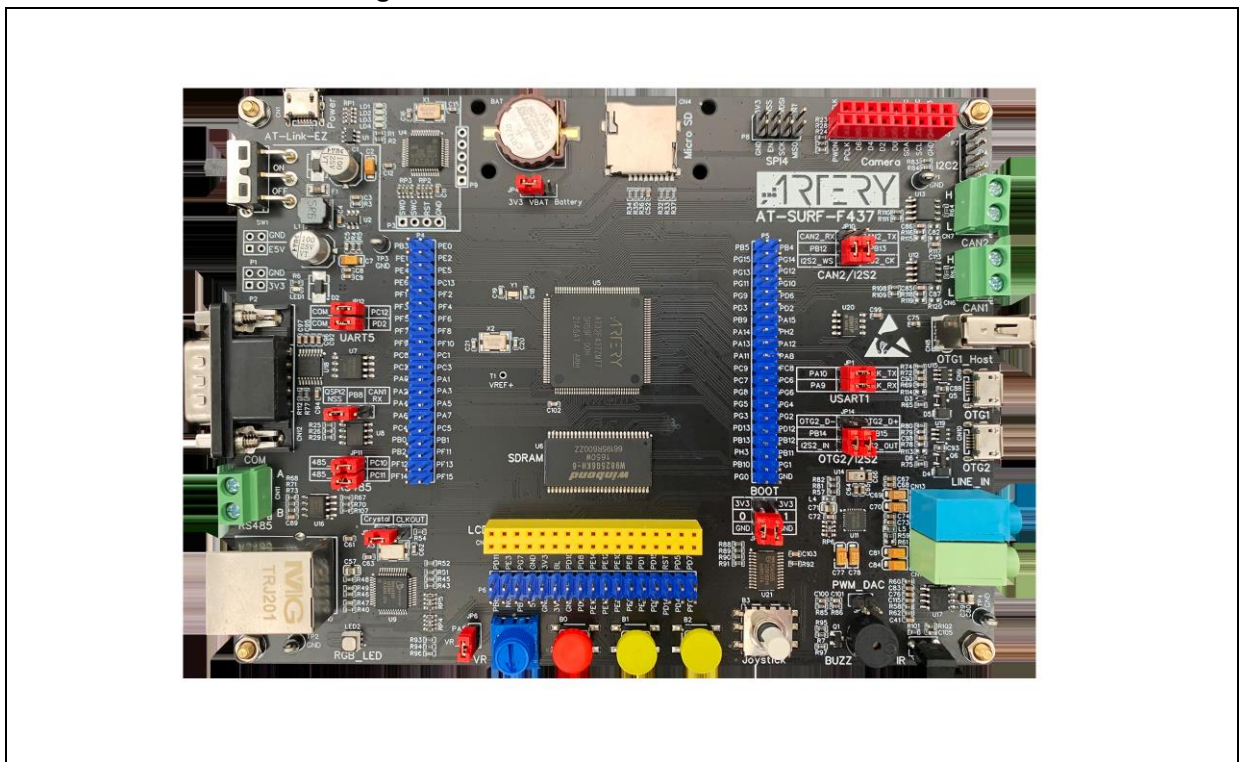


2 Quick start

2.1 Hardware resources

- 1) AT-START-F437 board
- 2) OV2640

Figure 1. AT-START-F437 V1.0 evaluation board



2.2 Software resources

- SC0120_SourceCode

Note: All of projects are built based on Keil 5. For the need to run in other compiling environments, user can make simple adjustments according to AT32xxx_Firmware_Library_V2.x.x\project\at_start_xxx\templates.

2.3 Example case

- 1) Connect the OV2640 to the Camera interface on the AT-SURF-F437 board
- 2) Open "SourceCode\SC0117_SourceCode_V2.0.0\utilities\uvc_camera_dvp_ov2640\mdk_v5"
- 3) Compile and download code to the board
- 4) Connect USB OTG1 to PC
- 5) Check the image information collected through the Camera software on the PC

3 Revision history

Table 1. Document revision history

Date	Revision	Changes
2023.05.12	2.0.0	Initial release

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